

Project Log: Design Decision

Project: Smart Campus Network Optimization Platform

Log Date: September 21, 2026

Log Type: Decision / Architecture Design Review

What Changed

The Day 0 Protocol node (already reclassified as a deterministic, non-LLM node in the September 20, 2026 decision log) was detailed further. Its implementation and trigger flow were defined as follows:

- AI agent connection simplified: the AI agent's link to the Day 0 Protocol is now a single code execution block, not a multi-step AI-mediated exchange. That block calls the Flask Python backend, which in turn calls the initial configuration script and reads the YAML file holding all device credentials.
- Credentials YAML generation: the YAML credentials file is produced by a separate Python script rather than written by hand. That script collects the required inputs — number of devices and the management IP subnet to draw addresses from — through a GUI connected to the web server, then writes the completed YAML file for the initial configuration script to consume.
- Web server entry point: on startup, the web server now prompts the user with a yes/no question asking whether to apply the Day 0 Protocol. Answering yes routes the user directly into the credentials-collection GUI described above.
- Chat box entry point: the same flow is also reachable from the chatbot interface — typing “apply day zero protocol” in the chat box triggers the identical routing into the credentials-collection GUI, so both entry points converge on the same GUI and the same generated YAML file.

Reasoning / Notes

This adjustment is scoped only to the Day 0 Protocol node and does not affect the multi-vendor network decision recorded separately. The design keeps the Day 0 Protocol deterministic end-to-end, consistent with the September 20, 2026 decision to remove LLM judgment from safety-critical, credential-handling paths: the AI agent's role is reduced to invoking a fixed code execution block rather than reasoning about configuration content, and credential collection is handled through a structured GUI input (device count, management subnet) rather than free-text or model-generated values. Offering the same trigger through both the web server's startup prompt and the chat box gives the user two equivalent, low-friction ways to initiate the same deterministic pipeline without introducing a second implementation to maintain.

Impact on Project Plan

- Team 2's deliverables now include: the credentials-collection GUI (web-server-connected), the Python script that generates the YAML credentials file from that GUI's input, and the code execution block linking the AI agent/chat trigger to the Flask backend and initial configuration script.
- No change to the classification of Day 0 Protocol as a deterministic node, per the September 20, 2026 decision log.
- No change to Team 1 (Wi-Fi optimization) or Team 3 (network topology) scope; the Cisco-only, GNS3-built network scope from the September 12, 2026 decision log is unaffected by this log.

Status

Design detail recorded for the Day 0 Protocol node's trigger flow, GUI-based credential collection, and YAML generation. Implementation and integration testing against the Cisco-only GNS3 topology to be confirmed in a future log once built.